CHEMISTRY

SECTION II

TIME – 1 HOUR AND 45 MINUTES

Directions:

Section II has 7 questions and lasts 1 hour and 45 minutes.

You may use the available paper for scratch work and planning, but you must write your answers in the free-response booklet. Label parts (e.g., A, B, C) and sub-parts (e.g., i, ii, iii) as needed. Use a pencil or a pen with black or dark blue ink to write your responses.

For each question, show your work for each part in the space provided for that part in the free-response booklet. Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

A calculator is allowed in this section. You may use a handheld four-function, scientific, or graphing calculator, or the calculator available in this application. Reference information, including a periodic table and lists of equations and constants, is available in this application and can be accessed throughout the exam.

You may pace yourself as you answer the questions in this section, or you may use these optional timing recommendations:

Questions 1–3 are long free-response questions. It is suggested that you spend about 23 minutes on each of those questions. Questions 4–7 are short free-response questions. It is suggested that you spend about 9 minutes on each of those questions.

You can go back and forth between questions in this section until time expires. The clock will turn red when 5 minutes remain—**the proctor will not give you any time updates or warnings.**

Note: This exam was originally administered digitally. It is presented here in a format optimized for teacher and student use in the classroom.

During the AP Exam administration, students have access to reference information. To see the reference information for this course, please visit AP Central:
https://apcentral.collegeboard.org/courses/ap-chemistry/exam

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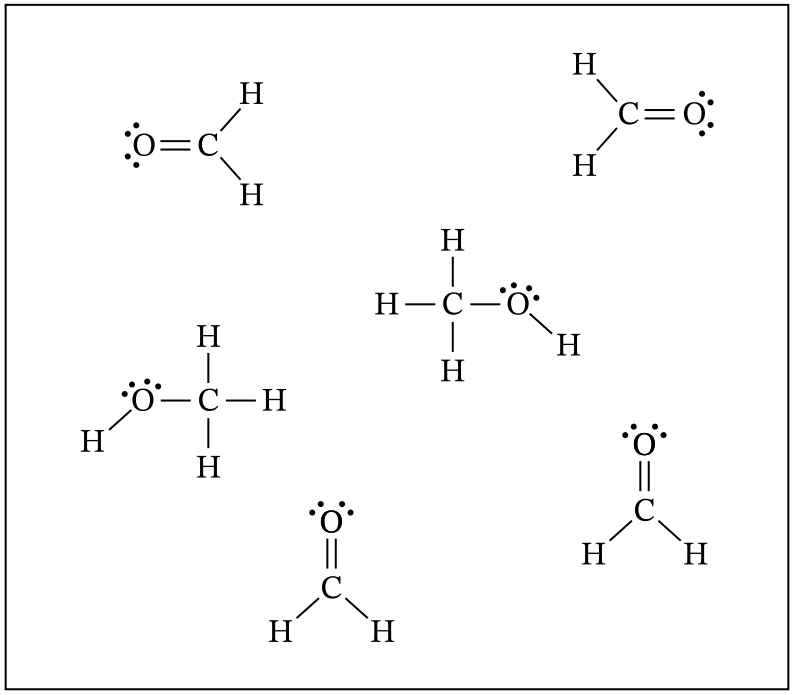
4. A scientist is investigating the properties of a mixture of CH_3OH and H_2CO. The scientist generates the following diagram to represent the mixture.



A. Identify the hybridization of the valence orbitals of the C atom in the H_2CO molecule.

B. In the diagram provided, draw a SINGLE dashed line (----) to represent a strong hydrogen-bonding attraction between one CH_3OH molecule and one H_2CO molecule in the mixture.

C. The scientist plans to cool a gaseous mixture of CH_3OH and H_2CO to form a liquid mixture and finds data on the two compounds. The data are summarized in the table.

Substance	Melting Point (K)	Boiling Point (K)	Enthalpy of Vaporization (kJ/mol)
CH_3OH	176	338	37.6
H_2CO	181	254	24.2

i. Propose a temperature to which the mixture should be cooled such that CH_3OH and H_2CO will both be liquids.

ii. The scientist analyzes the mixture after it is cooled and determines that 8.59 g of $CH_3OH(l)$ is present. Calculate the amount of thermal energy, in kJ, that was removed to condense the 8.59 g of CH_3OH (molar mass 32.04 g/mol) at its boiling point.

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